Performance Testing

AI/ML and Generative AI Track • Project Documentation

Project	Trip Genius AI (VoyagePlan Travel Planner)	**Date**	26 October 2026
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College	LNCTS	**University**	RGPV

Step 1. Testing overview

Field	Details
Testing tool	Manual production smoke testing in the browser (Console / Network). A load-testing tool such as k6, Locust, JMeter or Postman is recommended for future measurement.
Type	Manual functional smoke test completed; load / stress testing not yet performed.
Target module	GET /health, POST /api/itineraries/generate and the saved-trip workflow.
Environment	Hosted production: Vercel frontend + Render FastAPI backend + Supabase.
Tester	Kumkum Thakur
Test date	26 October 2026

Step 2. Smoke test results (reported by project owner)

Test case	Result
Sign up / sign in works	PASS
Generate an itinerary successfully	PASS
Save a trip successfully	PASS
Saved trip appears in My Trips	PASS
Edit preferences loads saved values	PASS
Trip remains saved after refreshing the page	PASS
No critical errors in browser console	PASS
GET /health returns a healthy response	PASS

Step 3. Suggested load-test scenarios

No.	Scenario	Virtual users	Duration	Expected outcome
1	Health endpoint baseline	1	60 sec	Consistent healthy responses; record latency
2	Generation baseline	1	5–10 requests	Valid response or readable provider error; record latency
3	Concurrent generation	5 (start conservatively)	60 sec	No crashes; monitor rate limits and latency
4	Saved trip retrieval	5	60 sec	Authenticated users only see permitted records

Step 4. Quantitative results

Numeric load-test measurements have not been collected yet, so they are recorded as **Not measured** rather than estimated.

Metric	Target / baseline	Actual	Status
Average response time	Set after baseline run	Not measured	Pending
Maximum response time	Set after baseline run	Not measured	Pending
Throughput	Measure requests / second	Not measured	Pending
Error rate	Establish acceptable threshold	Not measured	Pending
CPU utilisation	Monitor Render metrics	Not measured	Pending
Memory utilisation	Monitor Render metrics	Not measured	Pending

Step 5. Observations

Production smoke checks passed, which establishes that the core workflow functions end to end. They do not describe load capacity. When load testing, note that Groq rate limits and Render cold starts can affect latency of the generation endpoint.

Additional recommended tests

Category	Test	Expected outcome
Input validation	Blank destination	Readable validation error; no unnecessary AI call
Date validation	End date before start date	Request rejected or corrected
AI integration	Provider rate limit or outage	Clear error; recoverable UI state
Ownership	Account A requests Account B's trip	Request denied
AI quality	Same trip, different interests	Meaningfully different itineraries
Responsive UI	Narrow mobile viewport	Planner and My Trips remain usable